

Descriptive Statistics

subset of population that represents population

Data exploration phase - or - EDA (Exploratory Data Analysis)

Descriptive stats is the first step to deal with SAMPLE. Typical operation that a data analyst, data scientist, business analyst prefer includes:

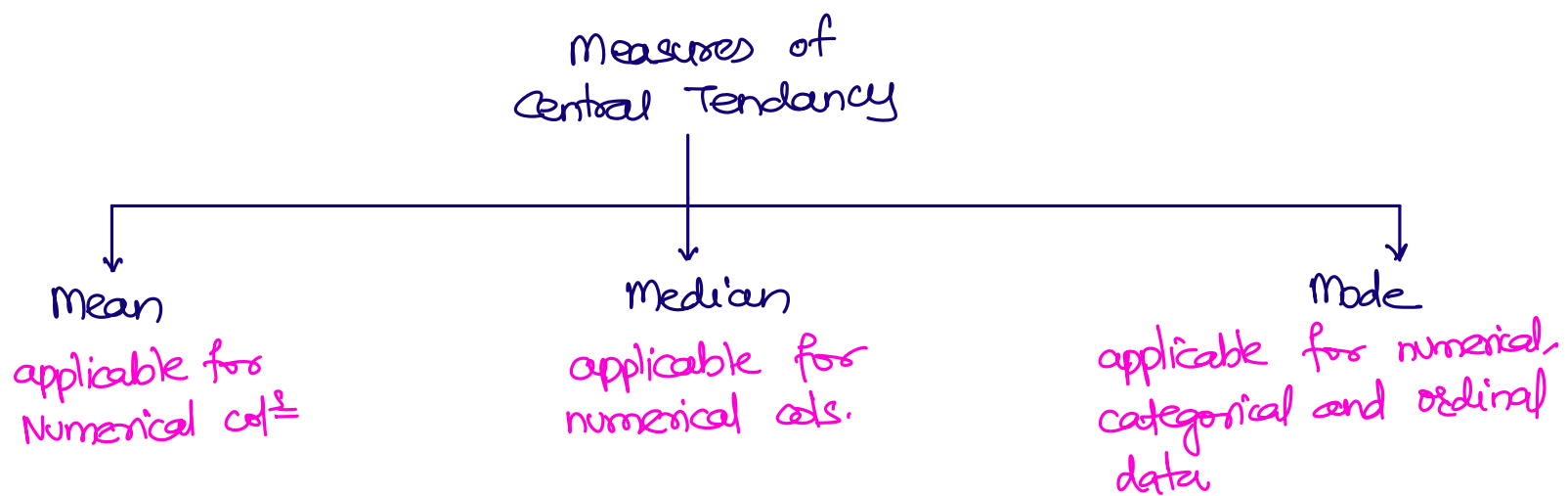
- ① Check the type of DISTRIBUTION for each numerical column.
- ② Identify and deal with INAPPROPRIATE data.
- ③ Identify and deal with OUTLIERS.
- ④ Identify and deal with CATEGORICAL data
- ⑤ Identify and deal with MISSING VALUES
- ⑥ Identify and deal with ORDINAL data
- ⑦ Discovers any

ASSOCIATION
RELATIONSHIP
PATTERN

between two or more variables (columns) in the given dataset.

Measures of Central Tendency

These are statistical tools used to **SUMMARIZE** the data based on **CENTRAL POINT** of the data.



① mean : The sum of all data points divided by the number of data pts.

$$\text{mean} = \frac{\sum_{i=1}^n x_i}{n}$$

$n \rightarrow$ no. of records in my dataset

eg: [2, 3, 4, 5]

$$\text{mean} = \frac{2+3+4+5}{4} = 3.5 //$$

② Median: Its the middle value of the data column; when its sorted from smallest to largest (ascending order)

calculating median:

- ① sort the data values in ascending order.
- ② Identify the middle position using:

a) if no. of records are odd; the median value is at $\left(\frac{n+1}{2}\right)^{\text{th}}$ position.

b) if no. of records are even; the median value is the avg. of $\left(\frac{n}{2}\right)^{\text{th}}$ pos. value & $\left(\frac{n}{2} + 1\right)^{\text{th}}$ position value.

eg: [3, 1, 4, 1, 5]

→ [1, 1, 3, 4, 5]

Here since no. of records is odd (5)

$\left(\frac{5+1}{2}\right)^{\text{th}}$ position value

3rd position = 3

eg: [3, 1, 4, 1]

→ [1, 1, 3, 4]

avg $\left(\left(\frac{4}{2}\right)^{\text{th}}$ pos value & $\left(\frac{4}{2} + 1\right)^{\text{th}}$ pos value $\right)$

2nd pos

3rd pos

$$\text{median} = \frac{1 + 3}{2} = 2$$

② mode → most frequent data | data with highest frequency
(count)

steps:

① count the frequency of each unique value.

② Identify the value that has the highest frequency.

eg: [1, 2, 2, 3, 3, 3, 4]

1 → 1

2 → 2

3 → 3

4 → 1

∴ mode = 3

eg: [1, 2, 2, 2, 3, 3, 3]

1 → 1

2 → 3

3 → 3

} → [2, 3] → ^{mode} 2 //

Note: if any/all unique values has same frequency, in that case, SORT the selected values in ASC order and return first value

[7, 7, 7, 7, 7, 7, 2, 2, 2, 2, 2, 2]

1 → 1

2 → 3

3 → 2

7 → 3

21 → 3

7, 21, 2

↓

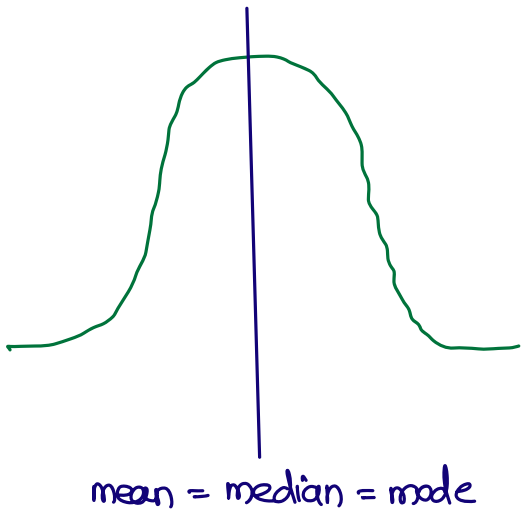
2, 7, 21

mode = 2

Data Distribution

This concept is applicable only for NUMERICAL columns.

① Normal Distribution | Gaussian Distribution | Bell Curve



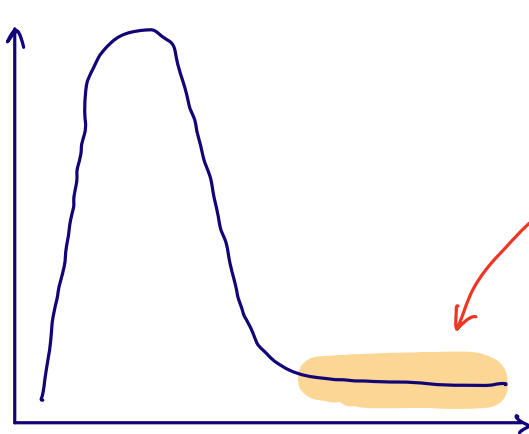
if the given data column forms a bell curve, chances are, given column is a GENERALIZED column.

data column successfully represents POPULATION

} High Probability

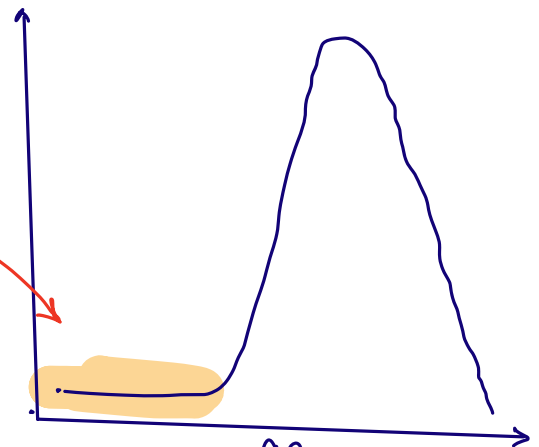
② Skew Distribution

graph with a tail



(a)
Right skew distribution

tail
-or-
extreme values
-or-
outliers



(b)
Left skew distribution

Quartiles | Quartiles

(column)
↑

Quartiles are the values that divide the dataset variable into 4 equal parts.

25%

Q_1

first
quartile

50%

Q_2

second
quartile

75%

Q_3

third
quartile

100%

Q_4

fourth
quartile

Consider the following dataset variable

$$D = \{12, 15, 7, 23, 10, 8, 14, 18, 5, 20\}$$

To calc quartiles for given data D

Step 1: Arrange the data in ascending order

$$\{ \underset{1}{5}, \underset{2}{7}, \underset{3}{8}, \underset{4}{10}, \underset{5}{12}, \underset{6}{14}, \underset{7}{15}, \underset{8}{18}, \underset{9}{20}, \underset{10}{23} \}$$

Step 2: Identify the median value of the above } Q_2
calculating Q_2

$$\text{median} = \frac{12+14}{2} = 13$$

$$\boxed{Q_2 = 13}$$

Step 3: Calc the first Quartile (Q_1)

Q_1 is the median of the lower half of the dataset

$$\boxed{Q_1 = 8}$$

step 4: Calculate third quartile (Q_3)

Q_3 is the median of the upper half of the dataset

$$Q_3 = 18$$

step 5: Calculate the fourth quartile (Q_4)

Q_4 is the max value of the dataset variable

$$Q_4 = 23$$

Calc Quantiles for

{ 5, 7, 8, 10, 12, 18, 14, 15, 20 }



$$Q_2 = 12$$

Q_1 is lower half median

$$\frac{7+8}{2} = 7.5$$

$$Q_4 = 20$$

Q_3 is upper half median

$$\frac{15+18}{2} = 16.5$$

Outliers

Outliers are extreme values that affect the **GENERAL TONE/NATURE** of the domain. To identify the outliers in dataset variable, you can use

TUKEY'S METHOD

— or —

1.5 IQR Rule

Inter Quartile Range

Algo

① Calc IQR for the given variable

$$IQR = Q3 - Q1$$

② Calc valid range for given variable

$$\text{lowerRange} = Q1 - (1.5 * IQR)$$

$$\text{upperRange} = Q3 + (1.5 * IQR)$$

